

ECON 1550: International Finance

# Money, Interest Rates, and Exchange Rates

# A model of the money market

| Exogenous variables |              | Endogenous variables |                        |                   |                       |
|---------------------|--------------|----------------------|------------------------|-------------------|-----------------------|
| Variable            | Description  | Variable             | Description            | Equation          | Type of equation      |
| $Y$                 | Real income  | $R$                  | Domestic interest rate | $M^d/P = L(R, Y)$ | Behavioral equation   |
| $M^s$               | Money supply | $M^d$                | Money demand           | $M^d = M^s$       | Equilibrium condition |
| $P$                 | Price level  |                      |                        |                   |                       |

# Description of the Money Market

- There are only two assets: money and domestic bonds
- Money
  - Can be used for transactions
  - Pays no interest
- Bonds
  - Cannot be used for transactions
  - Pay interest  $R \geq 0$

# Money Demand

- Money demand is higher when:
  - Higher price level  $P \rightarrow$  need more money to buy goods
  - Lower nominal interest rate  $R \rightarrow$  bonds less attractive
  - Higher income  $Y \rightarrow$  want more money to buy more
- Capture idea with a behavioral equation

$$M^d = P \times L\left(\underset{(-)}{R}, \underset{(+)}{Y}\right)$$

# Real money demand

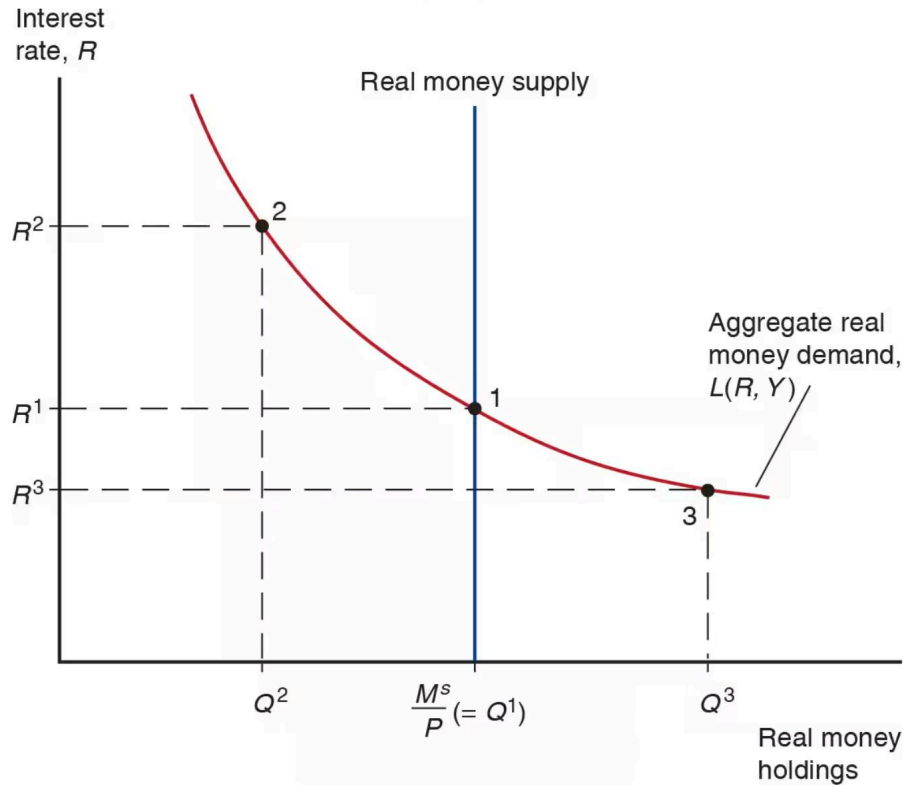
- Convenient to write in terms of real money demand

$$\frac{M^d}{P} = L(R, Y)$$

and real money supply

$$\frac{M^s}{P}$$

# Equilibrium in money market



# Money is defined by its function

- Medium of exchange
- Store of value
- Unit of account

# Many types of money (1/2)

- Commodity money: a physical commodity (like gold) that is used as money
- Convertible paper money: a piece of paper that can be exchanged for a commodity



## Many types of money (2/2)

- Fiat money: issued by a central bank, not backed by any commodity
- Digital currency: not backed by any commodity, privately issued, electronic payments

# Fiat money

1. Central bank liabilities are special because they are the unit of account.
2. Currency is a promise to deliver future central bank liabilities.

| Assets  | Liabilities | Assets       | Liabilities | Assets    | Liabilities |
|---------|-------------|--------------|-------------|-----------|-------------|
| \$0     | \$0         | \$1 currency | \$1 money   | \$1 bonds | \$1 money   |
| $t = 0$ |             | $t = 1$      |             | $t = 2$   |             |

# Monetary policy

- Deposits at the Fed are called reserves
- “The” interest rate is just interest on reserves

# Short-run FX and money market model

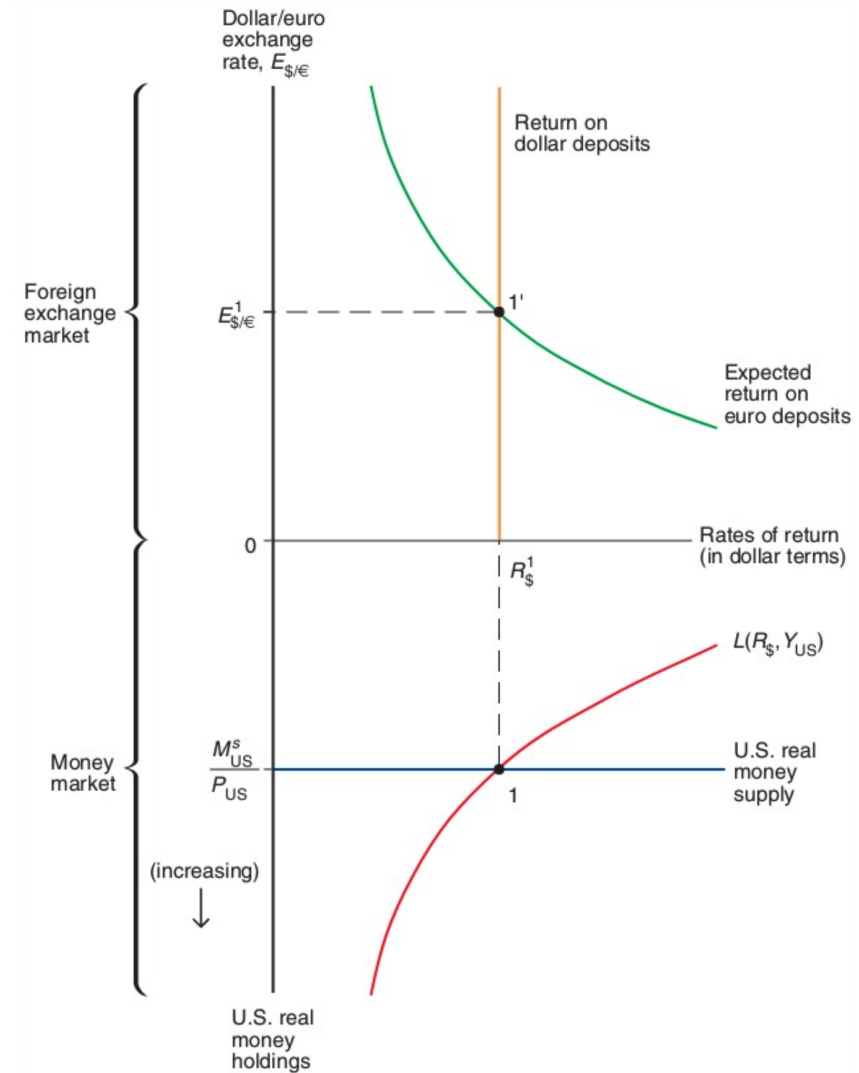
## Exogenous variables

| Variable | Description            |
|----------|------------------------|
| $R^*$    | Foreign interest rate  |
| $E^e$    | Expected exchange rate |
| $Y$      | Real income            |
| $M^s$    | Money supply           |
| $P$      | Price level            |

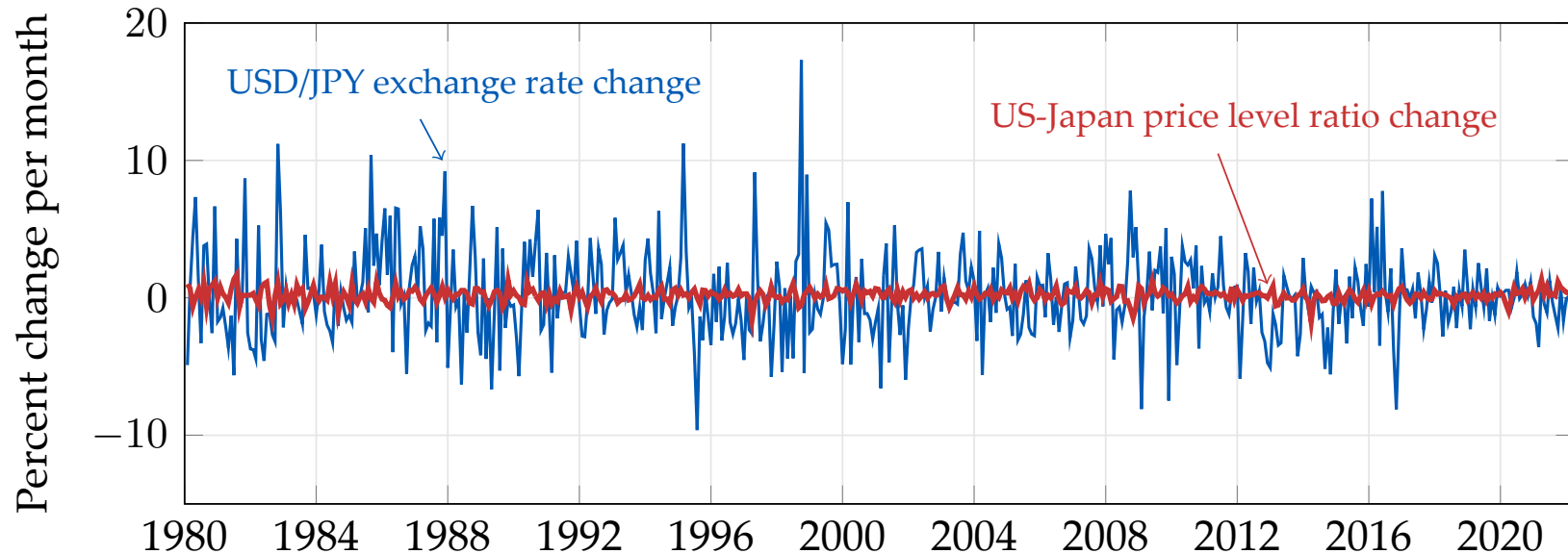
## Endogenous variables

| Variable | Description            | Equation                      | Type of equation      |
|----------|------------------------|-------------------------------|-----------------------|
| $E$      | Exchange rate          | $R = R^* + \frac{E^e - E}{E}$ | Equilibrium condition |
| $R$      | Domestic interest rate | $M^d/P = L(R, Y)$             | Behavioral equation   |
| $M^d$    | Money demand           | $M^d = M^s$                   | Equilibrium condition |

# Short-run equilibrium in FX and money markets



# Exchange rates are very volatile



Source: [FRED/CPIAUCSL](#); [FRED/CPALCY01JPM661N](#); [FRED/DEXJPUS](#) (end-of-month, inverted to USD/JPY).

# Conceptual definition of long-run equilibrium

- Hypothetical equilibrium that would result if economy runs indefinitely with no shocks
- Equivalently, the hypothetical equilibrium that would occur if prices were perfectly flexible and always adjusted instantaneously and frictionlessly

# Definition of the long run in the model

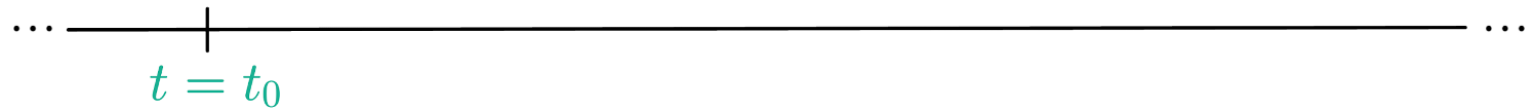
1. In the long run,  $E = E^e$ 
  - By definition of the long run, all variables constant
  - By assumption of rational expectations, expectations must be correct in the long run
  - In the model, it means  $E_{LR} = E_{LR}^e$



# Assumptions about the short run

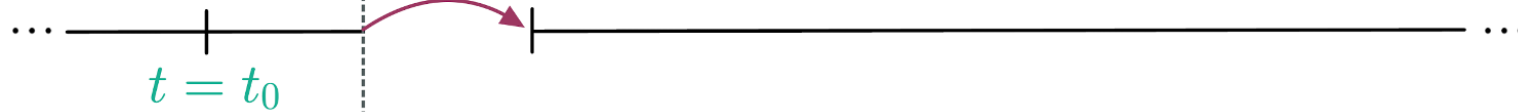
1. In the short run,  $P$  is fixed
  - Because prices are “sticky”
  - In the model, it means  $P_0 = P_{SR}$
2. In the short run,  $E^e$  equals long-run  $E$ 
  - By assumption of rational expectations
  - In the model, it means  $E_{SR}^e = E_{LR}$

Initial long run  
equilibrium



Initial long run  
equilibrium

Short run



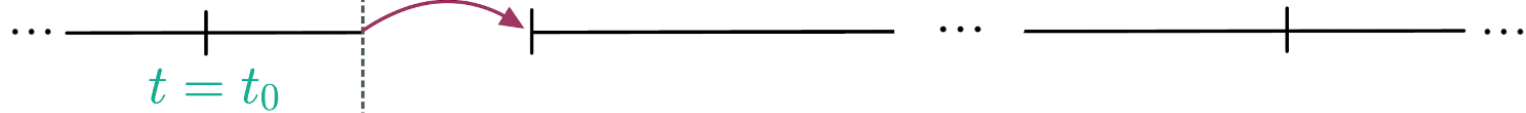
Exogenous  
variable  
changes

A red arrow points from the text "Exogenous variable changes" to the vertical dashed line.

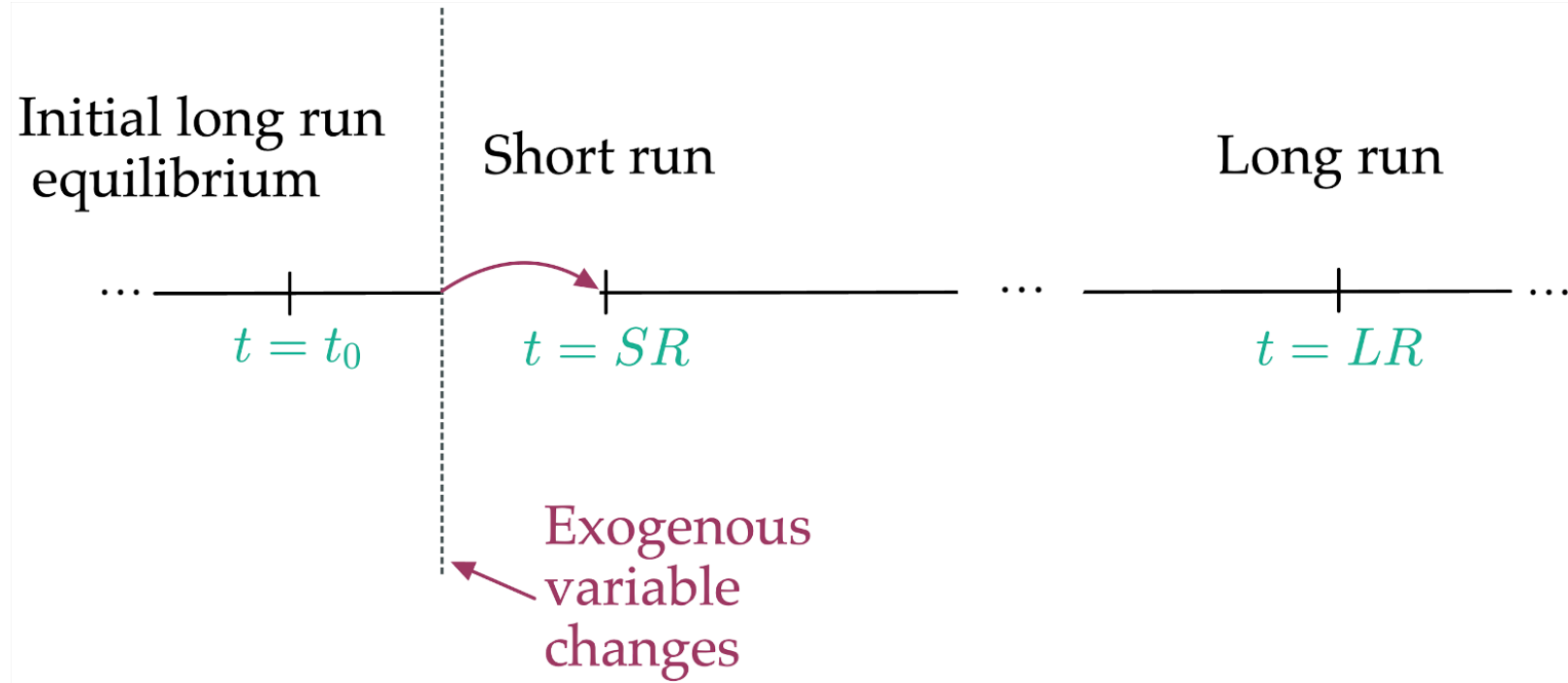
Initial long run  
equilibrium

Short run

Long run



Exogenous  
variable  
changes



# Results in the long run (1/3)

1. Long-run interest rate  $R_{LR}$  from FX market
  - Using  $E_{LR} = E_{LR}^e$  and UIP:

$$R_{LR} = R^* + \frac{E_{LR}^e}{E_{LR}} - 1$$
$$\Rightarrow R_{LR} = R^*$$

## Results in the long run (2/3)

2. Long-run price level  $P_{LR}$  from the money market
- Using  $R_{LR} = R^*$  and money market equilibrium condition:

$$\frac{M^s}{P_{LR}} = L(R_{LR}, Y) = L(R^*, Y)$$
$$\Rightarrow P_{LR} = \frac{M^s}{L(R^*, Y)}$$

# Results in the long run (3/3)

3. Long-run exchange rate  $E_{LR}$  from PPP (purchasing power parity)

- Because  $E_{LR}$  is a nominal price, in the long run it moves proportionally to the price level:

$$E_{LR} \propto P_{LR}$$



# Results in the short run (1/2)

1. Short-run exchange rate  $E_{SR}$  from UIP with  $E^e = E_{LR}$ 
  - Using UIP and the short-run assumptions:

$$R = R^* + \frac{E_{LR} - E_{SR}}{E_{SR}} \quad \Rightarrow \quad E_{SR} = \frac{E_{LR}}{1 + R - R^*}$$

# Results in the short run (2/2)

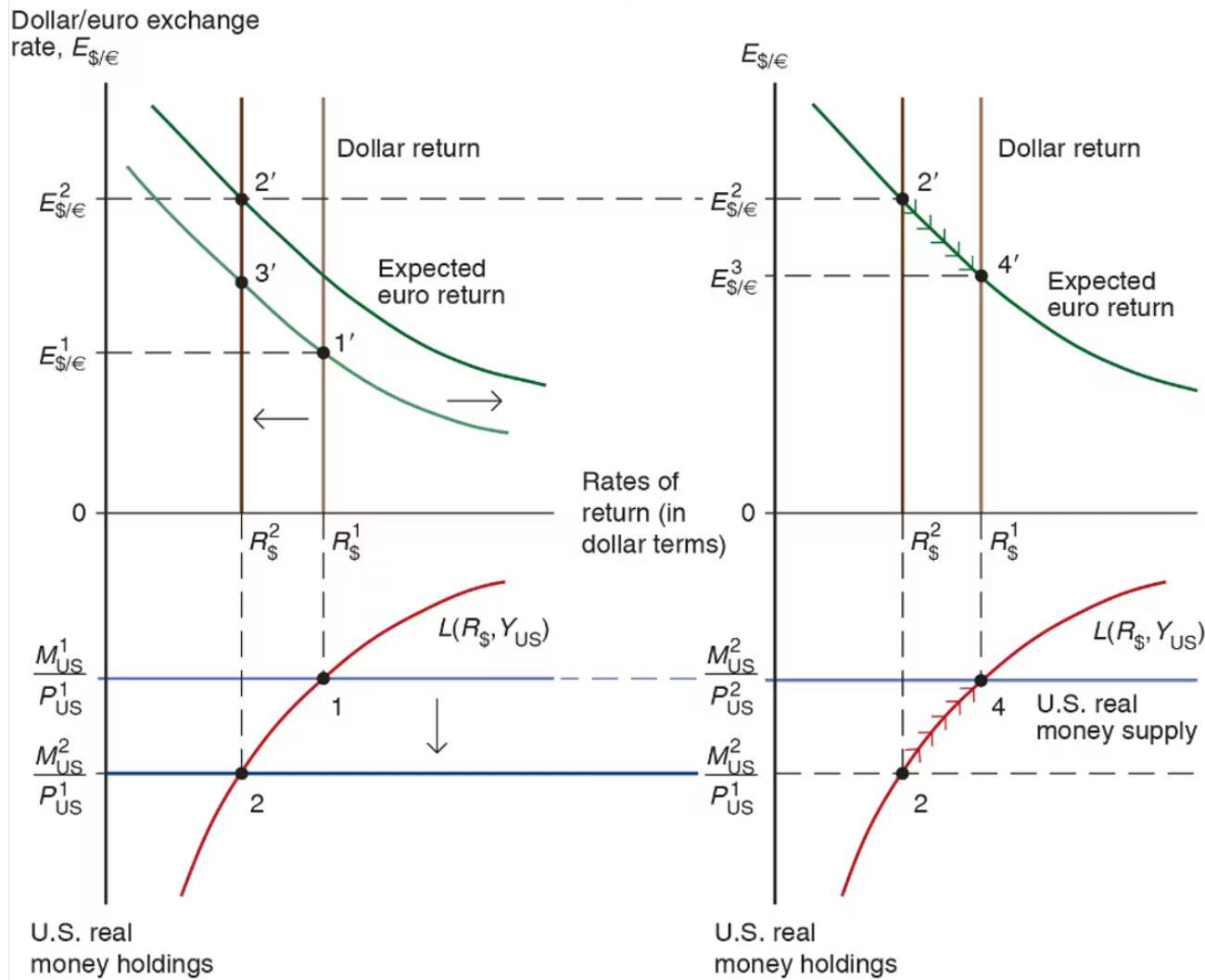
## 2. Short-run domestic interest rate $R_{SR}$ from the money market

- Using money market equilibrium condition and  $P$  fixed at  $P_0$  in the short run:

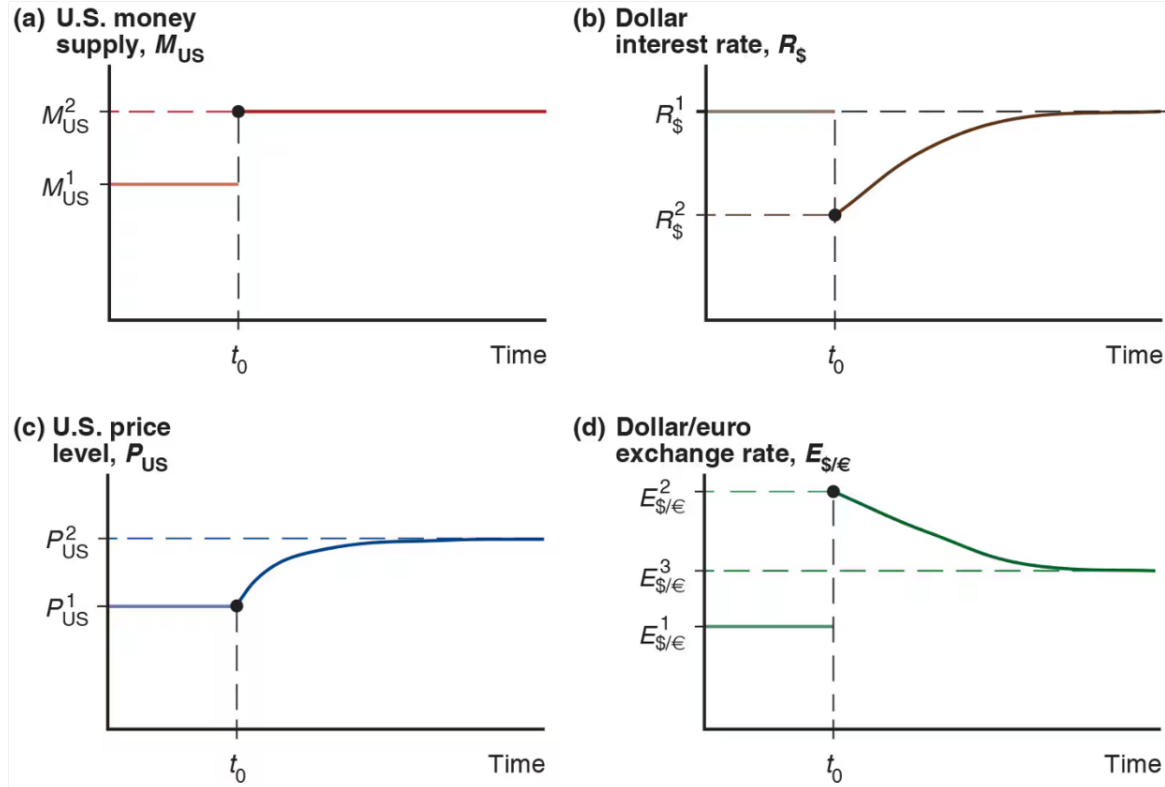
$$\frac{M^s}{P_{SR}} = L(R_{SR}, Y) \quad \Rightarrow \quad \frac{M^s}{P_0} = L(R_{SR}, Y),$$

which can be solved for  $R_{SR}$

# Permanent Money Supply Changes



# Exchange Rate Overshooting



# Long-run FX and money market model

## Exogenous variables

| Variable | Description           |
|----------|-----------------------|
| $Y^n$    | Potential output      |
| $M^s$    | Money supply          |
| $R^*$    | Foreign interest rate |

## Endogenous variables

| Variable | Description            | Equation                      | Type of eq    |
|----------|------------------------|-------------------------------|---------------|
| $E$      | Exchange rate          | $R = R^* + \frac{E^e - E}{E}$ | Eq. condition |
| $M^d$    | Money demand           | $M^d/P = M^s/P$               | Eq. condition |
| $E^e$    | Expected exchange rate | $E^e = E$                     | Behavioral    |
| $R$      | Domestic interest rate | $P = \frac{M^d}{L(R, Y^n)}$   | Eq. condition |
| $P$      | Price level            | P fixed in the short run      | Behavioral    |

# Overshooting

1. Properties of initial, SR, LR equilibria
2. Dynamic FX and money market model
3. Example
  - Solve with equations
  - Solve graphically

# Properties of initial, SR, and LR equilibria

Long run:

- $E^e = E$

$$\Rightarrow E_0^e = E_0 \text{ and } E_{LR}^e = E_{LR}$$

- $E$  proportional to  $P$  and  $M^s$

$$\Rightarrow E_0 = kM_0^s \text{ and } E_{LR} = kM_{LR}^s \quad (k \text{ is a parameter})$$

# Properties of initial, SR, and LR equilibria

Short run:

- Prices are fixed

$$\Rightarrow P_{SR} = P_0$$

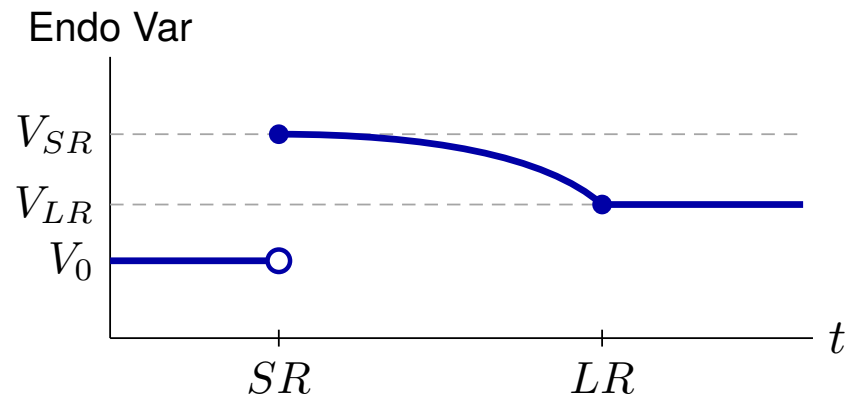
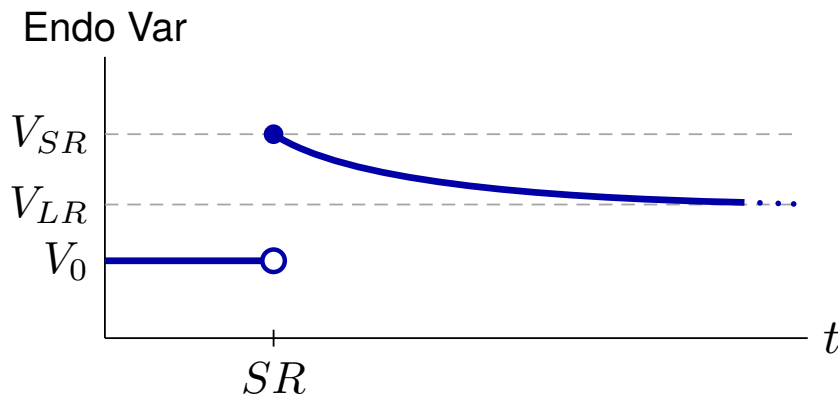
- Rational expectations

$$\Rightarrow E_{SR}^e = E_{LR}$$



# Transition between SR and LR

- Slow monotonic transition between SR value and LR value
- No indication of shape



# Dynamic FX and money market model

## Exogenous variables

| Variable | Description                                         |
|----------|-----------------------------------------------------|
| $M^s$    | Path of money supply                                |
| $R^*$    | Path of foreign interest rate                       |
| $Y$      | Path of income (GNP)                                |
| $k$      | Long-run proportionality constant for $E$ and $M^s$ |

## Endogenous variables

| Variable | Description            |
|----------|------------------------|
| $E$      | Exchange rate          |
| $E^e$    | Expected exchange rate |
| $P$      | Price level            |
| $R$      | Domestic interest rate |
| $M^d$    | Money demand           |

# Equations that apply at all times

$$\text{(UIP)} : R = R^* + \frac{E^e}{E} - 1 \quad \text{(equilibrium condition)}$$

$$\text{(MD)} : \frac{M^d}{P} = L\left(R, \underset{(-)}{Y}\right) \quad \text{(behavioral)}$$

$$\text{(MS = MD)} : \frac{M^s}{P} = \frac{M^d}{P} \quad \text{(equilibrium condition)}$$

# Equations that apply in SR equilibria only

(Sticky P) :  $P_{SR} = P_0$

(behavioral)

(RE) :  $E_{SR}^e = E_{LR}$

(behavioral)

# Equations that apply in LR equilibria only

$$\text{(PPP) or (Flex P)} : E_0 = kM_0^s, \quad E_{LR} = kM_{LR}^s \quad \text{(behavioral)}$$

$$\text{(Defn LR) or (RE)} : E_0^e = E_0, \quad E_{LR}^e = E_{LR} \quad \text{(behavioral)}$$

# Example

Need to specify:

1. Function  $L(R, Y)$
2. Path of exogenous variables

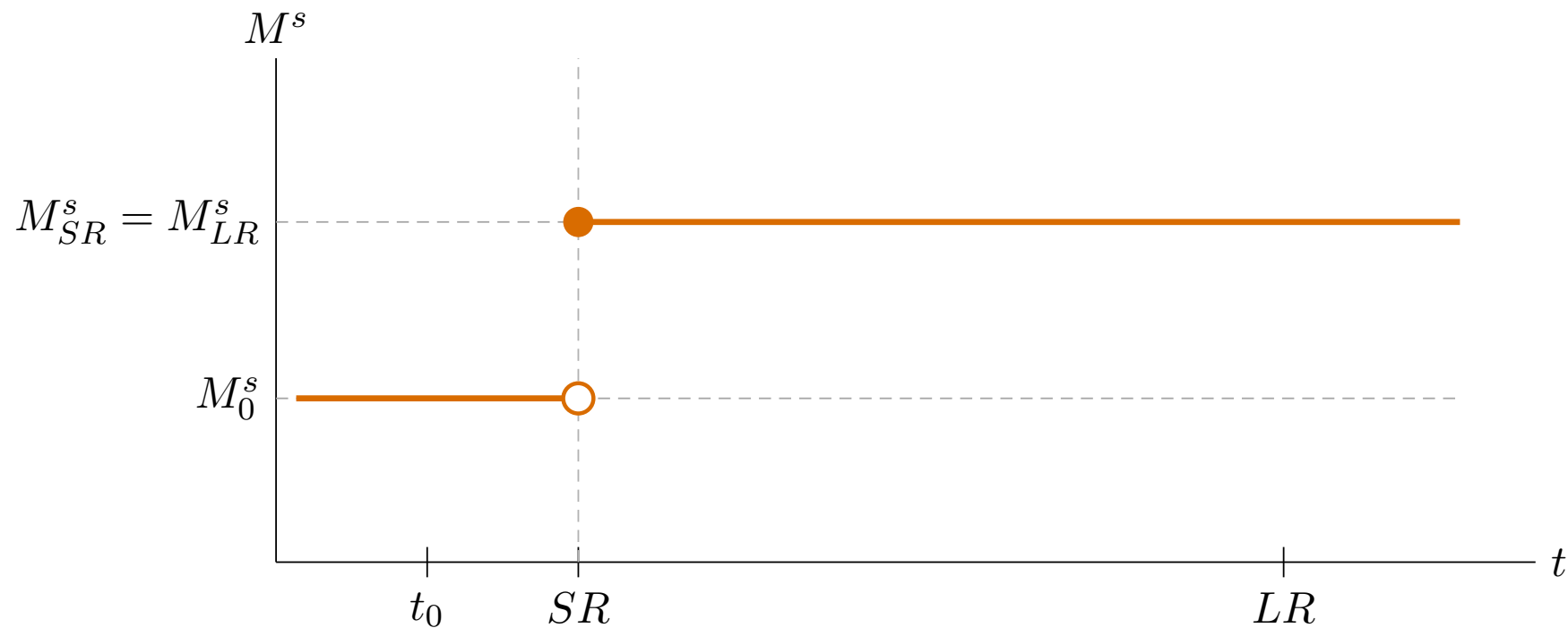
Assume:

1.  $L(R, Y) = \frac{Y}{1 + R}$

2.  $R^* = 0.08, Y = 1, k = 1$

$$\{M_0^s, M_{SR}^s, M_{LR}^s\} = \{1, 1.05, 1.05\}$$

# Example: Permanent increase in $M^s$



# Solution steps

1. Solve initial LR equilibrium ( $t = t_0$ )
2. Solve LR equilibrium ( $t = LR$ )
3. Solve SR equilibrium ( $t = SR$ )



# Solution sub-steps

## LR (steps 1 and 2)

- (a) Flex P / PPP  $\longrightarrow E$
- (b) Defn LR / RE  $\longrightarrow E^e$
- (c) UIP  $\longrightarrow R$
- (d) MS = MD  $\longrightarrow P$

## SR (step 3)

- (a) P fixed  $\longrightarrow P$
- (b) MS = MD  $\longrightarrow R$
- (c) RE  $\longrightarrow E^e$
- (d) UIP  $\longrightarrow E$

# Step 1: Solve initial LR equilibrium ( $t = t_0$ )

$$(a) \quad \text{Flex P / PPP} \rightarrow E \quad : \quad E_0 = kM_0^s = 1 \cdot 1 = 1$$

$$(b) \quad \text{Defn LR / RE} \rightarrow E^e \quad : \quad E_0^e = E_0 = 1$$

$$(c) \quad \text{UIP} \rightarrow R \quad : \quad R_0 = R^* + \frac{E_0^e}{E_0} - 1 \\ = 0.08 + \frac{1}{1} - 1 = 0.08$$

# Step 1: Solve initial LR equilibrium ( $t = t_0$ )

$$\begin{aligned} \text{(d) } MS = MD \rightarrow P & \quad : \quad \frac{M_0^s}{P_0} = \frac{Y_0}{1 + R_0} \\ & \Rightarrow P_0 = M_0^s \cdot \frac{1 + R_0}{Y_0} \\ & \quad = 1 \times 1.08 = 1.08 \end{aligned}$$

Solution for  $t_0$ :  $P_0 = 1.08$ ,  $R_0 = 0.08$ ,  $E_0^e = 1$ ,  $E_0 = 1$

## Step 2: Solve LR equilibrium ( $t = LR$ )

(a) Flex P / PPP  $\rightarrow E$  :  $E_{LR} = kM_{LR}^s = 1 \cdot 1.05 = 1.05$

(b) Defn LR / RE  $\rightarrow E^e$  :  $E_{LR}^e = E_{LR} = 1.05$

(c) UIP  $\rightarrow R$  :  $R_{LR} = R_{LR}^* + \frac{E_{LR}^e}{E_{LR}} - 1$   
 $= 0.08 + \frac{1.05}{1.05} - 1 = 0.08$

## Step 2: Solve LR equilibrium ( $t = LR$ )

$$\begin{aligned} \text{(d) MS=MD} \rightarrow P & \quad : \quad \frac{M_{LR}^s}{P_{LR}} = \frac{Y_{LR}}{1 + R_{LR}} \\ & \Rightarrow P_{LR} = M_{LR}^s \cdot \frac{1 + R_{LR}}{Y_{LR}} \\ & = 1.05 \times 1.08 = 1.134 \end{aligned}$$

Solution for LR:  $P_{LR} = 1.134$ ,  $R_{LR} = 0.08$ ,  $E_{LR}^e = 1.05$ ,  $E_{LR} = 1.05$

## Step 3: Solve SR equilibrium ( $t = SR$ )

$$(a) \quad P \text{ fixed} \rightarrow P \quad : \quad P_{SR} = P_0 = 1.08$$

$$(b) \quad MS=MD \rightarrow R \quad : \quad \frac{M_{SR}^s}{P_{SR}} = \frac{Y_{SR}}{1 + R_{SR}}$$
$$\Rightarrow \frac{1.05}{1.08} = \frac{1}{1 + R_{SR}}$$
$$\Rightarrow 1 + R_{SR} = \frac{1.08}{1.05}$$
$$\Rightarrow R_{SR} = \frac{1.08}{1.05} - 1 \approx 0.02857$$

## Step 3: Solve SR equilibrium ( $t = SR$ )

$$(c) \quad RE \rightarrow E^e : \quad E_{SR}^e = E_{LR} = 1.05$$

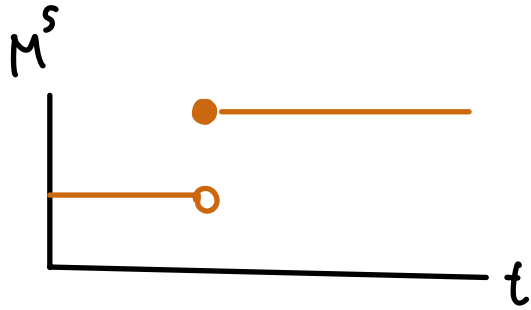
$$(d) \quad UIP \rightarrow E : \quad R_{SR} = R_{SR}^* + \frac{E_{SR}^e}{E_{SR}} - 1$$

$$\Rightarrow 0.02857 = 0.08 + \frac{1.05}{E_{SR}} - 1$$

$$\Rightarrow E_{SR} = \frac{1.05}{0.02857 - 0.08 + 1} = \frac{1.05}{0.94857} \approx 1.1069$$

Solution for SR:  $P_{SR} = 1.08$ ,  $R_{SR} \approx 0.029$ ,  $E_{SR}^e = 1.05$ ,  $E_{SR} \approx 1.107$

# Example: solution summary



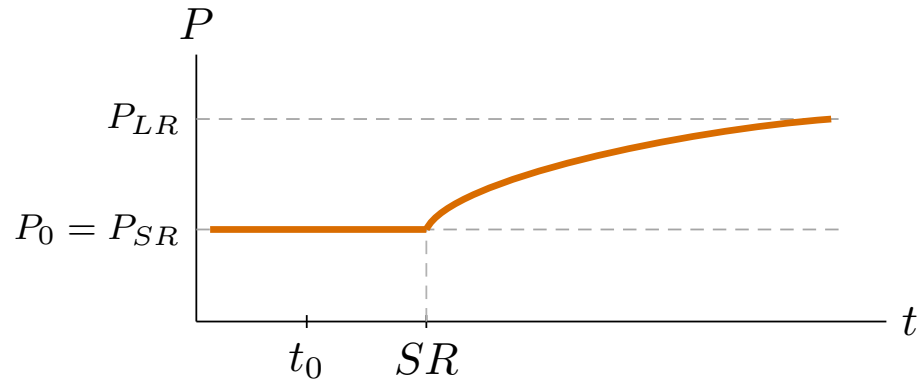
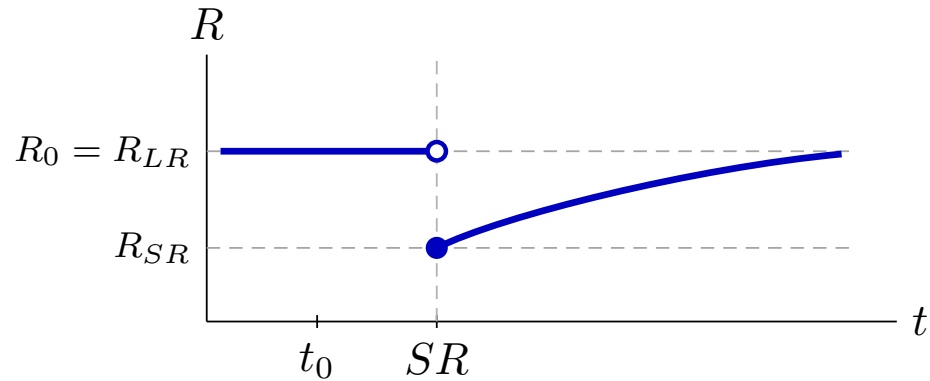
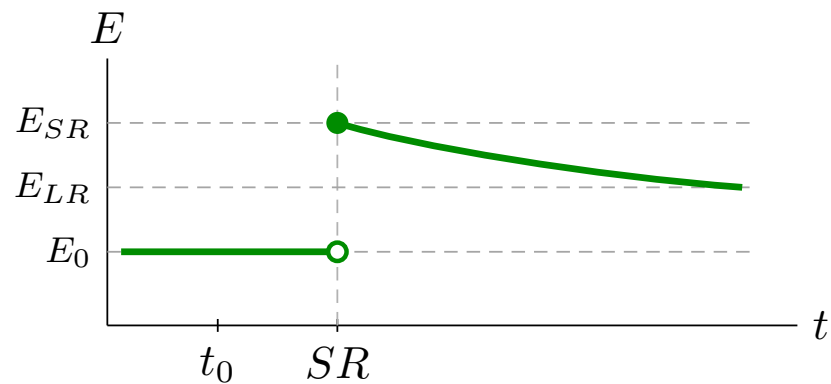
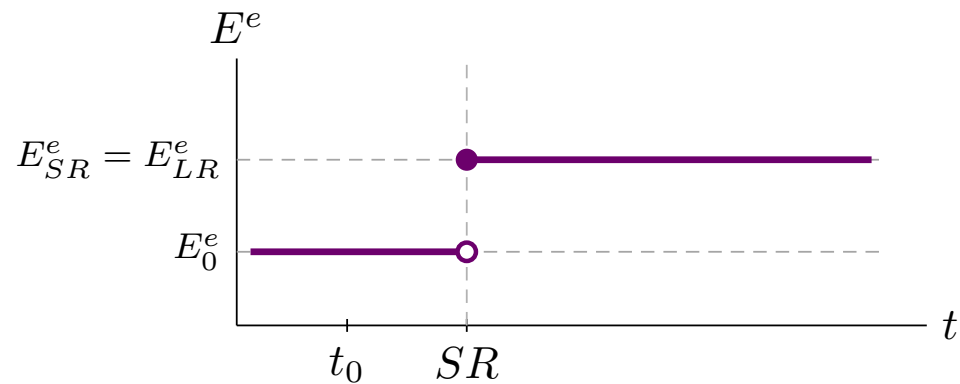
|       | $t_0$ | $SR$  | $LR$  |
|-------|-------|-------|-------|
| $P$   | 1.08  | 1.08  | 1.134 |
| $R$   | 0.08  | 0.029 | 0.08  |
| $E^e$ | 1     | 1.05  | 1.05  |
| $E$   | 1     | 1.107 | 1.05  |

$1.107 > 1.05$   
 $E_{SR} > E_{LR}$

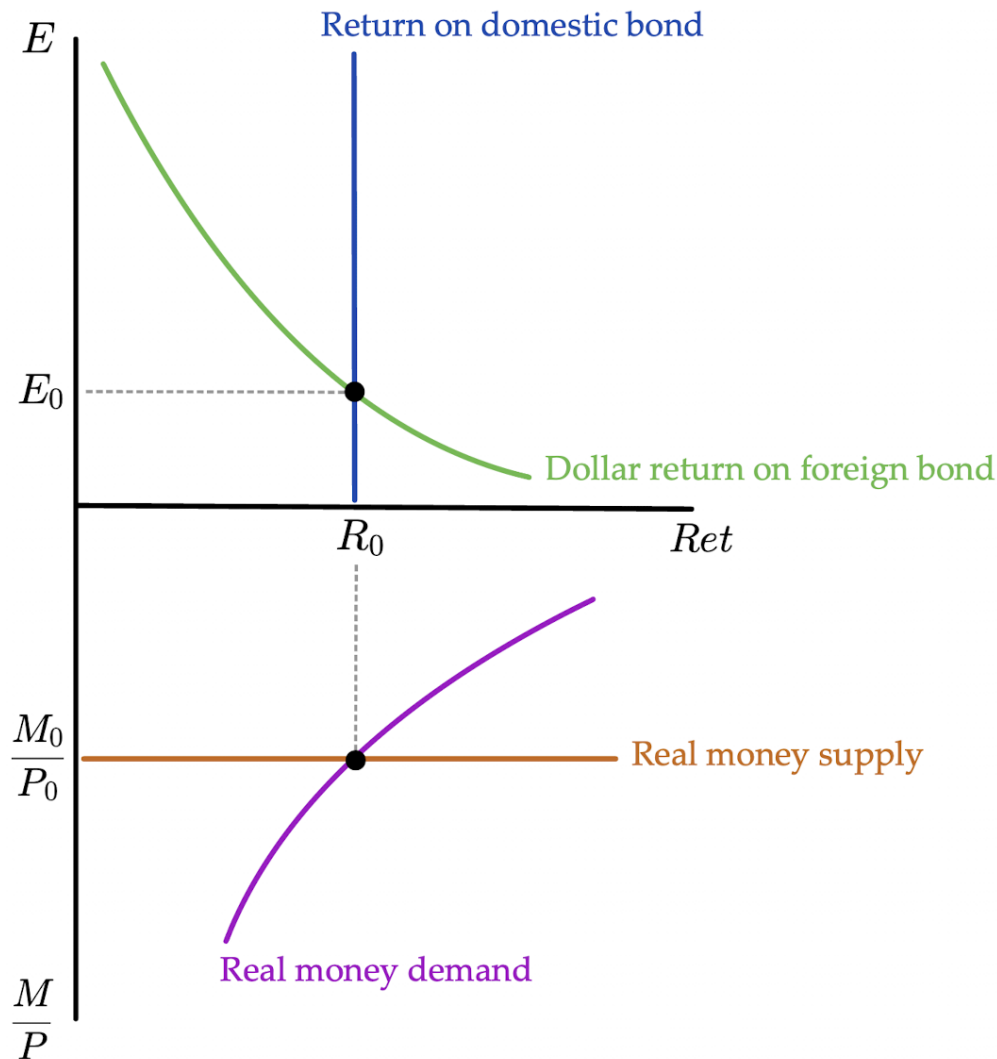
} OVERSHOOT



# Paths of solution



Solve with plots:  
Initial LR  
equilibrium

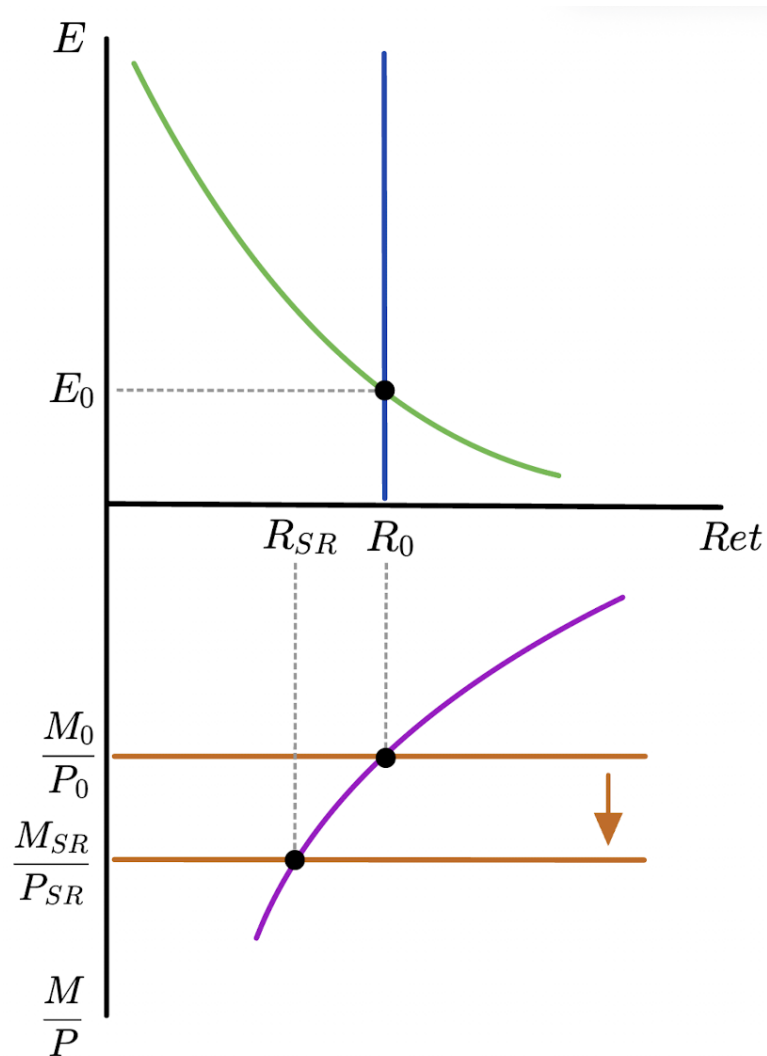


Higher  $M^s$   
lowers  $R$

$$M^s \uparrow$$

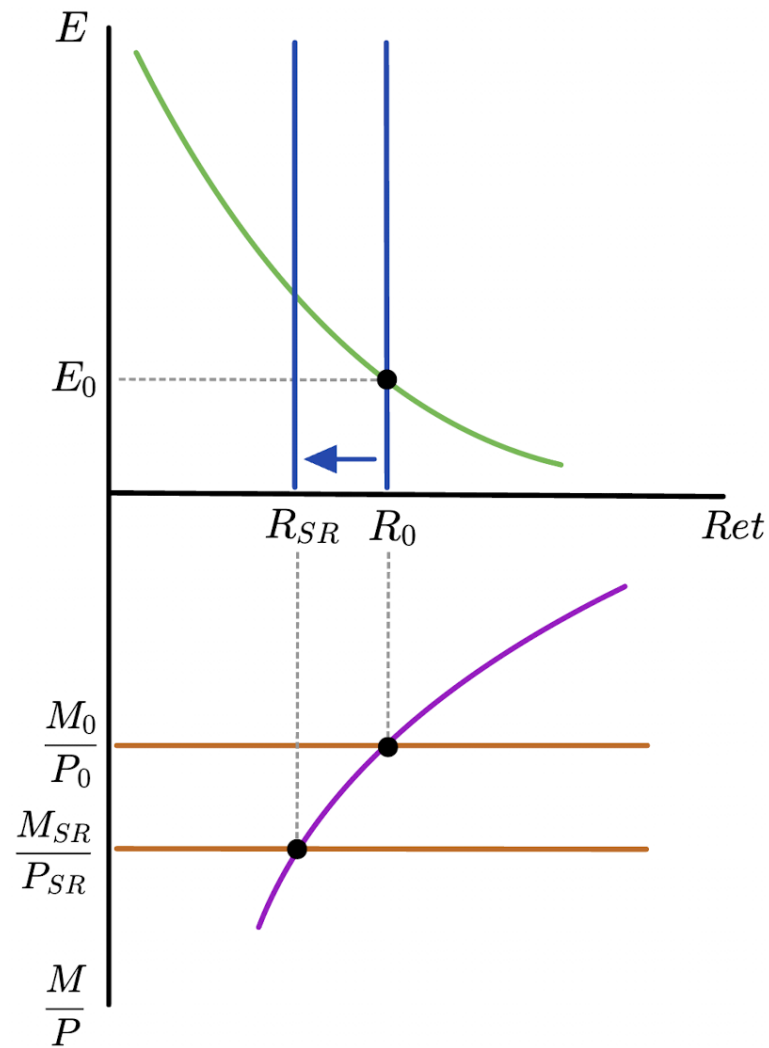
$$\frac{M^s}{P} \uparrow$$

$$R_{SR} \downarrow$$



Lower  $R$   
causes  
depreciation

$R \uparrow$



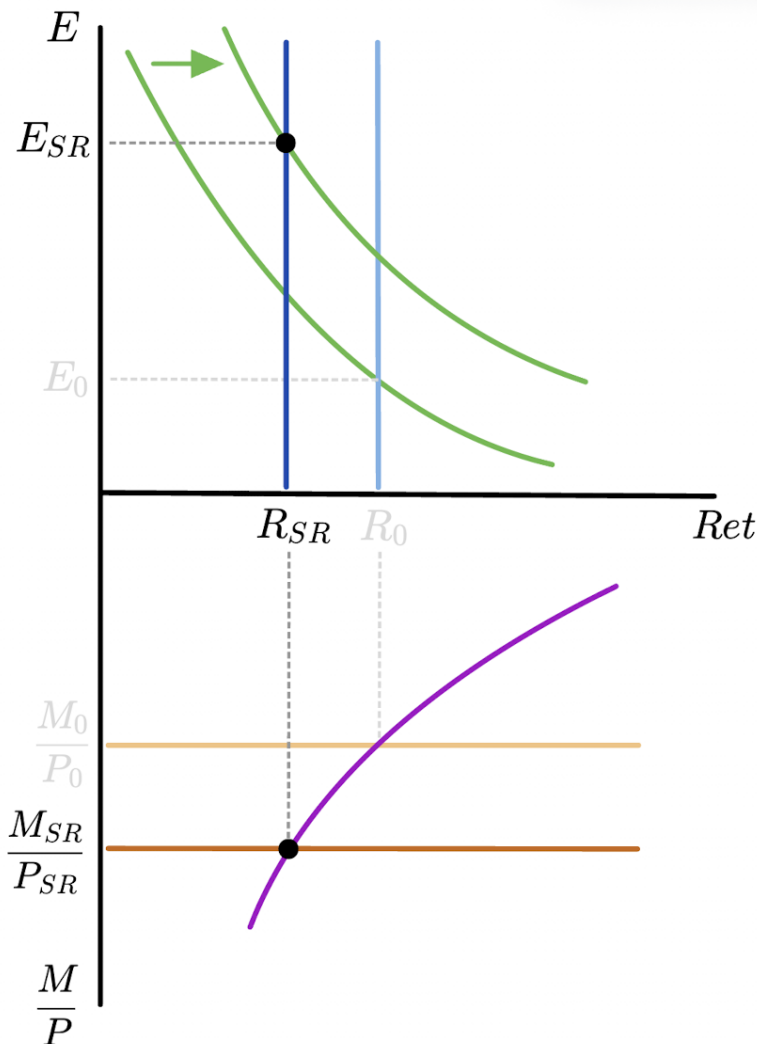
Higher  $E^e$   
shifts dollar  
return on  
foreign bond

$M^s \uparrow$  PERMANENTLY

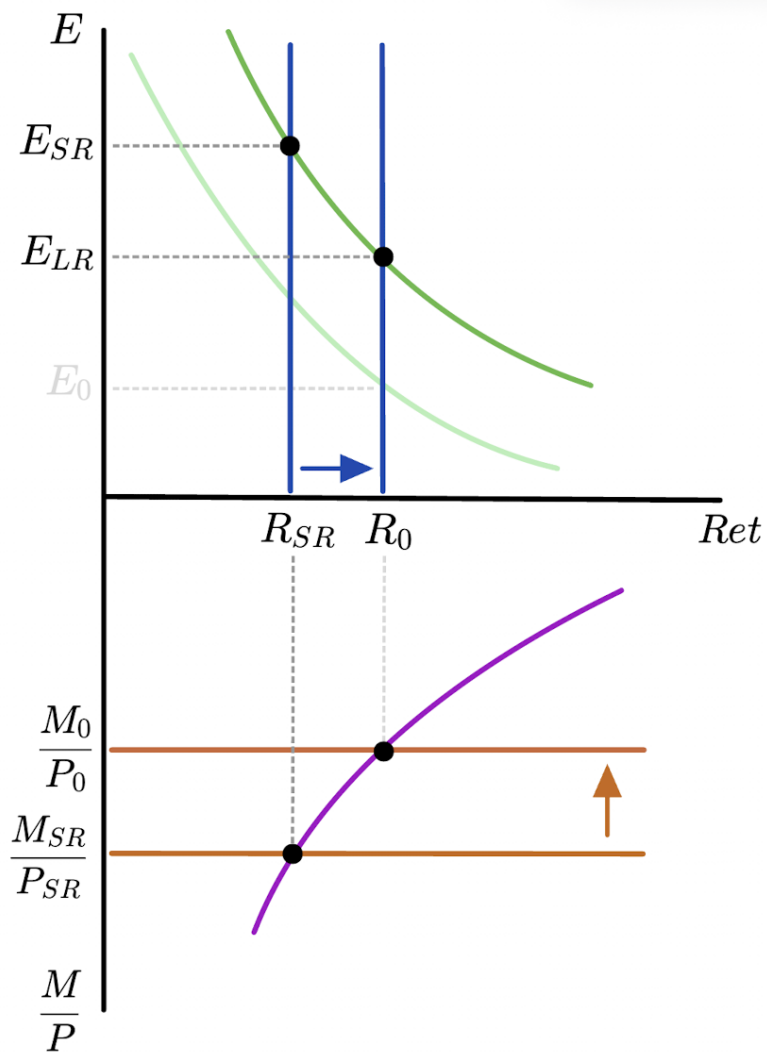
$\Rightarrow P \uparrow$  IN LR

$E \uparrow$  IN LR

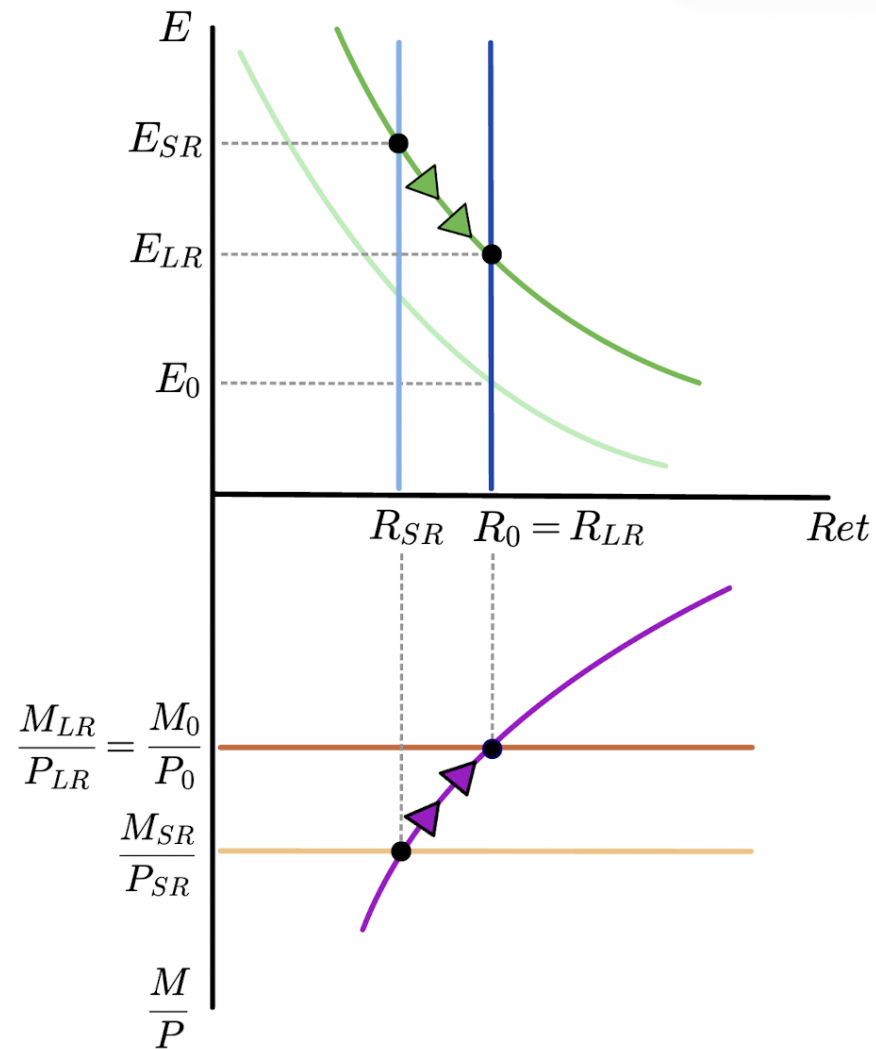
$E_{SR}^e \uparrow$



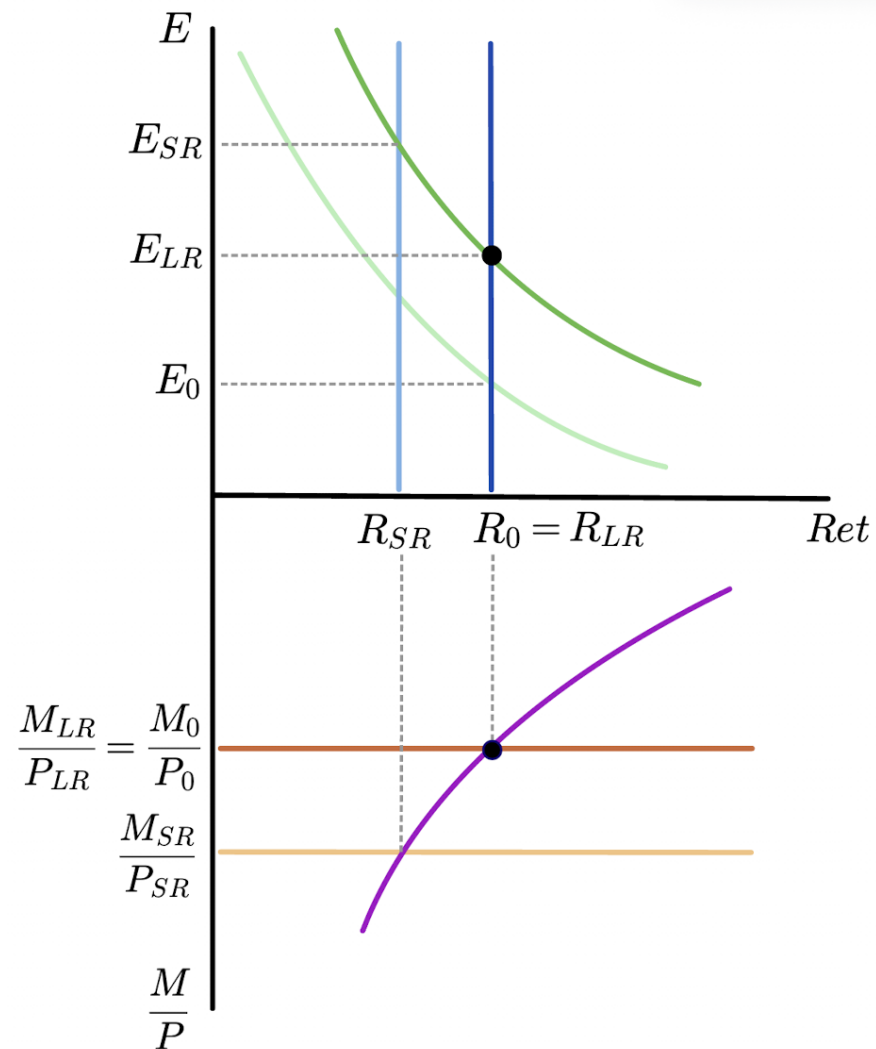
# SR equilibrium



# Transition between SR and LR

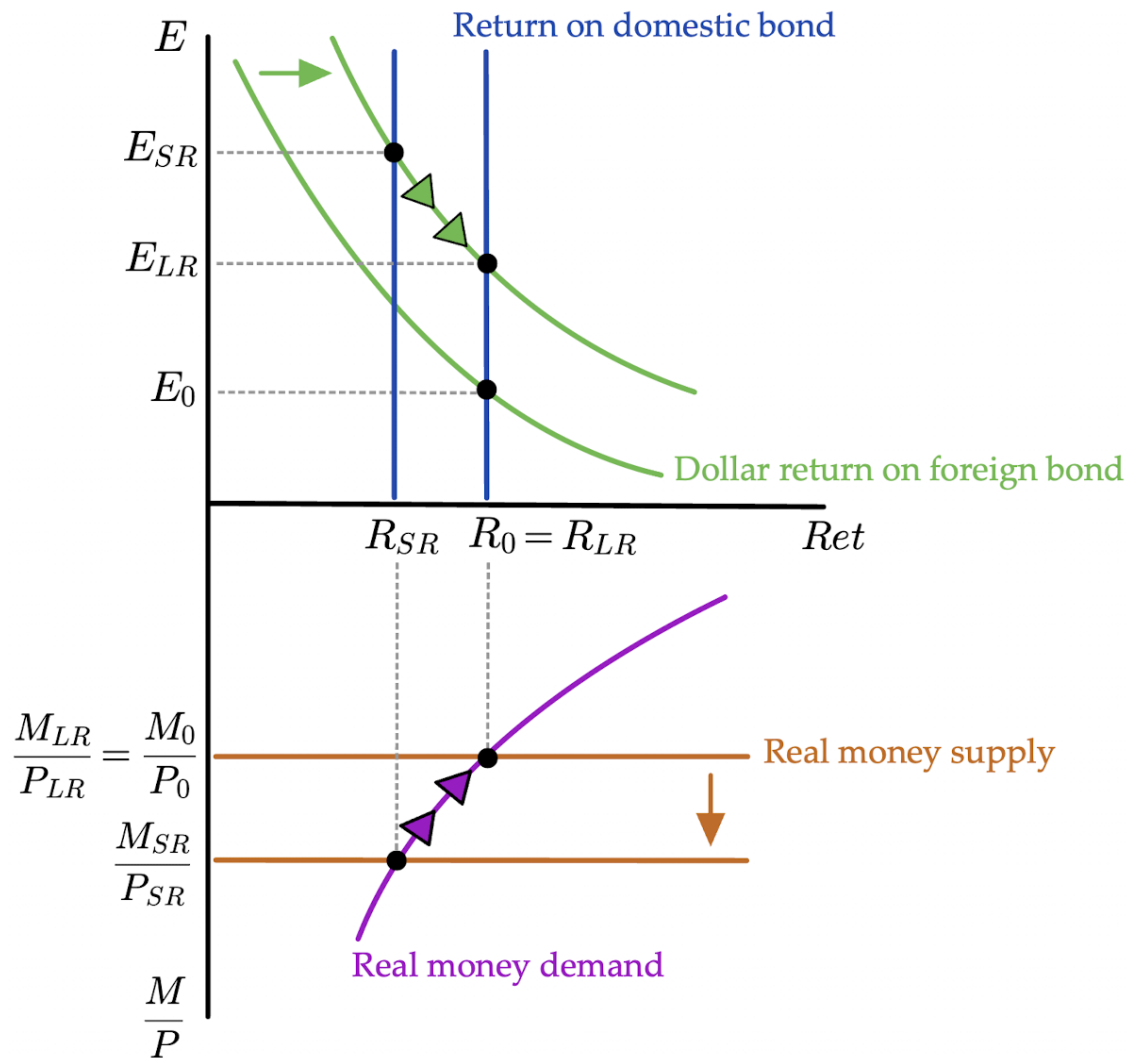


Final LR  
equilibrium





Solve with plots:  
All steps  
together



## Equilibrium Determination Map

|       | Initial      | SR        | LR           |
|-------|--------------|-----------|--------------|
| $E$   | flex P / PPP | FX mkt    | flex P / PPP |
| $E^e$ | defn LR      | RE        | defn LR      |
| $R$   | FX mkt       | money mkt | FX mkt       |
| $P$   | money mkt    | fixed P   | money mkt    |

PPP = purchasing power parity  $\Rightarrow E \propto P \propto M^s$

defn LR = definition of the long run  $\Rightarrow E_0^e = E_0$  and  $E_{LR}^e = E_{LR}$

RE = rational expectations  $\Rightarrow \begin{cases} E_{SR}^e = E_{LR} \\ E_0^e = E_0 \text{ and } E_{LR}^e = E_{LR} \end{cases}$